

Job Search Assistant

AI-Powered CV-Based Job Matching System with Deadline Notifications

Upload your CV, get matched with relevant jobs from multiple sources, and never miss a deadline!

Features

-  **Smart CV Parsing** - Extracts name, email, skills, experience from PDF/DOCX
 -  **Multi-Source Job Search** - Scrapes jobs from Bangladesh sites + global APIs
 -  **Intelligent Matching** - Scores jobs 0-100% based on your profile
 -  **Deadline Tracking** - Browser & email notifications for upcoming deadlines
 -  **Country-Specific** - Automatically searches relevant job sites based on your location
 -  **Job Categorization** - Excellent, Good, Fair, and Possible matches
 -  **Application Tracking** - Mark jobs as applied
 -  **Global Access** - Run locally, access globally via ngrok
-

Quick Start

Prerequisites

- Python 3.8+
- pip
- Internet connection

Installation

```
bash
```

```
# 1. Clone/Download the project
```

```
cd job-assistant
```

```
# 2. Install dependencies
```

```
pip install -r requirements.txt
```

```
python -m spacy download en_core_web_sm
```

```
# 3. Create .env file
```

```
echo "SECRET_KEY=your-secret-key" > .env
```

```
echo "REMOTEOK_API_ENABLED=true" >> .env
```

```
# 4. Run the app
```

```
python app.py
```

```
# 5. Open browser
```

```
# Go to: http://localhost:5000
```

That's it! See [QUICK_START.md](#) for detailed 5-minute setup.

Documentation

File	Description
QUICK_START.md	Get running in 5 minutes
API_SETUP_GUIDE.md	Get free API keys for more jobs
SETUP_INSTRUCTIONS.md	Complete setup & deployment guide
test_apis.py	Test your API keys

Supported Job APIs (All FREE!)

API	Free Limit	Coverage	Signup Required
RemoteOK	Unlimited*	Remote jobs worldwide	✗ No
Adzuna	500/month	Bangladesh, India, UK, US, etc.	✓ Yes
The Muse	500/day	Professional & tech jobs	✓ Yes
Reed	Unlimited	UK jobs	✓ Yes
JSearch	100/month	Global search	✓ Yes (RapidAPI)
BDJobs	Unlimited*	Bangladesh jobs	✗ No (scraping)
Chakri	Unlimited*	Bangladesh jobs	✗ No (scraping)

*With reasonable rate limits

Screenshots

1. Upload CV

Click to upload CV (PDF/DOCX) |

Maximum file size: 16MB |

2. CV Information

Name: John Doe

Email: john@example.com

Skills: Python, JavaScript, React, SQL

Experience: 3 years

Country: Bangladesh

3. Job Matches

Python Developer [95% Match] |

Tech Solutions BD |

 Dhaka, Bangladesh |

 Deadline: 2025-11-15 (28 days) |

[View Details] [Mark Applied] |

Project Structure

```
job-assistant/
├── app.py           # Main Flask application
├── requirements.txt  # Python dependencies
├── .env             # API keys & config (create this)
├── .gitignore       # Git ignore rules
├──
├── modules/         # Core functionality
│   ├── cv_parser.py # CV extraction logic
│   ├── job_scraper.py # Multi-source job scraping
│   ├── matcher.py   # CV-to-job matching algorithm
│   └── notifier.py  # Deadline notification system
├──
├── models/          # Database models
│   └── database.py  # User, Job, Notification models
├──
├── static/          # Frontend assets
│   ├── css/style.css # Styling
│   ├── js/main.js   # Frontend logic
│   └── uploads/      # CV storage
├──
├── templates/       # HTML templates
│   └── index.html    # Main interface
├──
└── data/            # Database storage
    └── job_assistant.db # SQLite database (auto-created)
```

Configuration

Minimum Configuration (.env)

```
bash

SECRET_KEY=your-secret-key-here
REMOTEOK_API_ENABLED=true
```

Full Configuration (.env)

```
bash
```

```
# Flask
```

```
SECRET_KEY=your-secret-key
```

```
FLASK_ENV=development
```

```
HOST=0.0.0.0
```

```
PORT=5000
```

```
# Job APIs
```

```
ADZUNA_APP_ID=your-app-id
```

```
ADZUNA_APP_KEY=your-app-key
```

```
THEMUSE_API_KEY=your-api-key
```

```
REED_API_KEY=your-api-key
```

```
RAPIDAPI_KEY=your-rapidapi-key
```

```
REMOTEOK_API_ENABLED=true
```

```
# Email Notifications
```

```
MAIL_USERNAME=your-email@gmail.com
```

```
MAIL_PASSWORD=your-app-password
```

```
MAIL_DEFAULT_SENDER=your-email@gmail.com
```

```
# Job Sites
```

```
BDJOBS_ENABLED=true
```

```
CHAKRI_ENABLED=true
```

```
PROTHOMALO_ENABLED=true
```

```
# Scraping
```

```
SCRAPING_DELAY=2
```

```
USER_AGENT=Mozilla/5.0...
```

See [API_SETUP_GUIDE.md](#) for how to get free API keys.



Testing

Test API Keys

```
bash
```

```
python test_apis.py
```

Output:

- ✓ Adzuna API: Working! - Found 150 jobs
- ✓ The Muse API: Working! - 45 pages available
- ✓ RemoteOK API: Working! - Found 1247 remote jobs
- ✗ Reed API: Failed - API key not found in .env
- ✗ RapidAPI (JSearch): Failed - API key not found in .env
- ✓ Email Config: Working! - Configured for john@gmail.com

SUMMARY: 4/6 services working

Test with Sample CV

1. Use your own CV (best results)
2. Ensure it has: skills, experience, email, phone
3. Format: PDF or DOCX

Global Access

Option 1: ngrok (Recommended)

```
bash

# Install ngrok from https://ngrok.com/download
ngrok http 5000

# Output:
# Forwarding: https://abc123.ngrok.io -> http://localhost:5000
```

Share the `https://abc123.ngrok.io` URL with anyone!

Option 2: Deploy to Cloud

Heroku:

```
bash

# Create Procfile
echo "web: gunicorn app:app" > Procfile

# Deploy
heroku create your-app-name
git push heroku main
```

PythonAnywhere: Upload files via dashboard

AWS/Azure/GCP: See `SETUP_INSTRUCTIONS.md`

1. CV Parsing

```
python
```

```
# Extracts:
```

- Name (using spaCy NER)
- Email (regex patterns)
- Phone (regex patterns)
- Skills (keyword matching)
- Experience (years extraction)
- Education (keyword detection)
- Country (location detection)

2. Job Scraping

```
python
```

```
# Sources:
```

- Adzuna API (500 requests/month)
- RemoteOK API (unlimited)
- The Muse API (500 requests/day)
- BDJobs.com (web scraping)
- Chakri.com (web scraping)
- More APIs as configured

3. Matching Algorithm

```
python
```

```
Score = (  
    Skills Match (50%) +  
    Experience Match (30%) +  
    Location Match (20%)  
)
```

```
Categories:
```

- Excellent: 80-100%
- Good: 60-79%
- Fair: 40-59%
- Possible: 30-39%

4. Notifications

python

Checks every 6 hours for:

- Deadlines **in next 3** days
- Sends browser notifications
- Sends email alerts (**if** configured)



Database Schema

sql

-- Users table

```
User {  
  id: Integer (PK)  
  name: String  
  email: String  
  phone: String  
  country: String  
  experience_years: Integer  
  skills: JSON  
  education: Text  
  cv_filename: String  
  created_at: DateTime  
}
```

-- Jobs table

```
Job {  
  id: Integer (PK)  
  user_id: Integer (FK)  
  title: String  
  company: String  
  location: String  
  job_url: String  
  deadline: DateTime  
  source: String  
  requirements: Text  
  match_score: Float  
  is_applied: Boolean  
  is_notified: Boolean  
}
```

-- Notifications table

```
Notification {  
  id: Integer (PK)  
  user_id: Integer (FK)  
  job_id: Integer (FK)  
  notification_date: DateTime  
  message: Text  
  is_sent: Boolean  
}
```



Security & Privacy

Data Protection

- ☒ API keys stored in `.env` (not committed to Git)
- ☒ Uploaded CVs stored locally (not shared)
- ☒ Database is local SQLite (not cloud)
- ☒ User data never leaves your server

Best Practices

```
bash

# 1. Add .env to .gitignore
echo ".env" >> .gitignore

# 2. Use strong SECRET_KEY
SECRET_KEY=$(python -c 'import secrets; print(secrets.token_hex(32))')

# 3. Change default passwords
# Never use default credentials in production

# 4. Enable HTTPS in production
# Use SSL certificates (Let's Encrypt is free)
```



Troubleshooting

Common Issues

1. "spaCy model not found"

```
bash

python -m spacy download en_core_web_sm
```

2. "No module named 'cv_parser'"

```
bash

# Create __init__.py files
touch modules/__init__.py
touch models/__init__.py
```

3. "Port 5000 already in use"

```
bash

# Change port in .env
PORT=5001
```

4. "No jobs found"

- Check internet connection
- Verify API keys with `python test_apis.py`
- Enable mock data (already included)
- Check console logs for errors

5. "CV parsing failed"

- Ensure CV is PDF or DOCX
- Check file size (max 16MB)
- Try with a different CV
- Check console for specific errors

6. "Database locked"

```
bash

# Close all running instances
# Delete and recreate database
rm data/job_assistant.db
python app.py
```

Performance Optimization

For Better Results

1. **Use specific skills** - "Python Django" vs "Programming"
2. **Update CV regularly** - More recent = better matches
3. **Enable multiple APIs** - More sources = more jobs
4. **Set realistic experience** - Don't inflate years

Server Optimization

```
python

# For production, use gunicorn:
pip install gunicorn
gunicorn -w 4 -b 0.0.0.0:5000 app:app

# Or waitress (Windows-friendly):
pip install waitress
waitress-serve --port=5000 app:app
```

Future Enhancements

Planned Features

- ☐ Auto-apply to jobs
- ☐ Cover letter generation (AI)
- ☐ Interview preparation tips
- ☐ Salary insights & negotiation
- ☐ LinkedIn integration
- ☐ Indeed API integration
- ☐ Multi-user support with authentication
- ☐ Mobile app (React Native)
- ☐ Resume builder
- ☐ Job alerts via SMS
- ☐ Analytics dashboard

Contributing

Want to add a feature? Fork and submit a PR!

License

This project is open source and available under the MIT License.

Acknowledgments

- **spaCy** - NLP and entity recognition
 - **Flask** - Web framework
 - **Beautiful Soup** - Web scraping
 - **Adzuna** - Job search API
 - **RemoteOK** - Remote jobs API
 - **The Muse** - Professional jobs API
-

Support

Get Help

-  Read documentation files
-  Run `python test_apis.py`
-  Check console logs for errors
-  Check email configuration

Resources

- [Flask Documentation](#)
 - [spaCy Documentation](#)
 - [Adzuna API Docs](#)
 - [ngrok Documentation](#)
-

Happy Job Hunting!

Built with  for job seekers everywhere.

Start finding your dream job today! 

Quick Links

-  [Quick Start Guide](#)
-  [API Setup Guide](#)
-  [Complete Setup Instructions](#)
-  [Test Your APIs](#)